

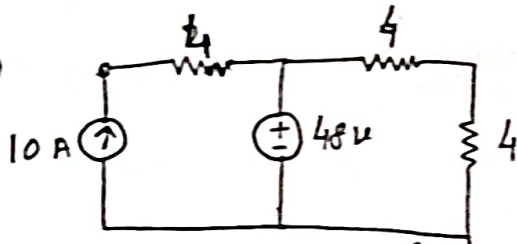
II B.E. EXAMINATION DEC- 2023
E & TC/ E & I ENGG.
3ETRG1/ 3EIRG1: NETWORK ANALYSIS

Duration: 3 Hrs.

Max. Marks : 60

Note: Attempt any two parts of each question. Each question carries equal marks. Make suitable assumptions if necessary.

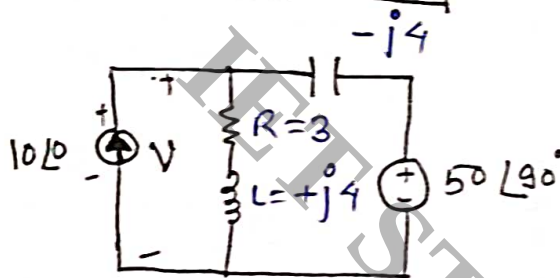
Q.1 (a)



Solve the network using (a) mesh analysis and (b) node analysis in Ideal condition.

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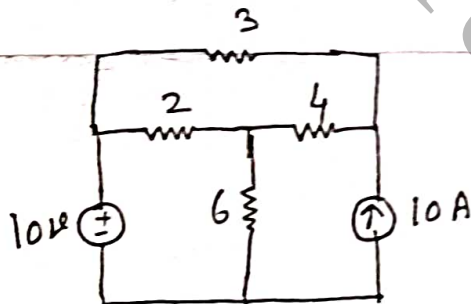
(b)



By Super-position theorem, find the voltage V in the given circuit.

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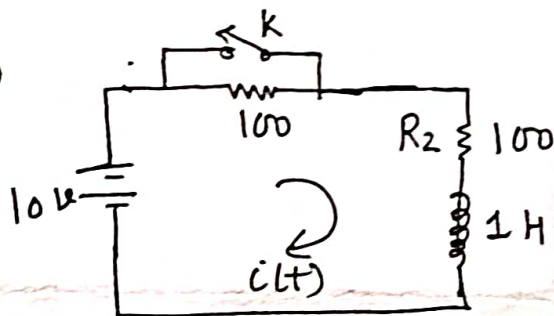
(c)



Find the current in 3Ω resistor using Norton's theorem.

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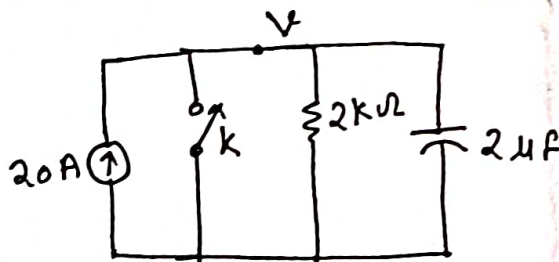
Q.2 (a)



In the given network, the switch K is closed at $t=0$. Find (i) $i(t)$, (ii) V_{R2} and (iii) V_L the voltage across the inductor, at $t=5\text{ msec}$

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(b)



Switch K is opened at $t=0$ with zero capacitor voltage. Find (i) V (ii) dV/dt and (iii) d^2V/dt^2 at $t=0+$.

6

$$\frac{d^2V}{dt^2}$$

- (c) Solve the equation if $\frac{d^2v}{dt^2} + 3\frac{dv}{dt} + 2v = 10 \sin 10t$ for $v(0+) = 1$ and $dv/dt(0+) = -1$ 6

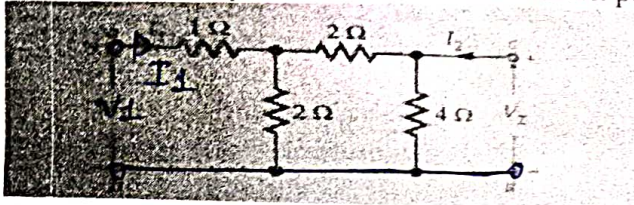
- Q. 3** (a) Check the stability of the network functions whose characteristic equations are given by: 6

(i) $s^4 + 11s^3 + 25s^2 + 3s + 2 = 0$

(ii) $s^5 + 3s^4 + 3s^3 + 4s^2 + s + 1 = 0$

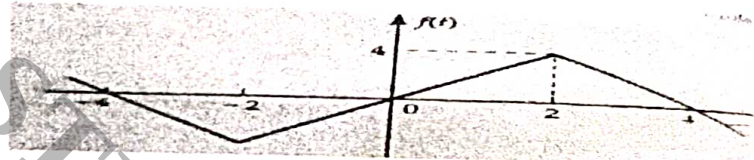
Also find no. of roots to left side of, on the and to the right side of, imaginary axis in each case.

- (b) Determine the Z parameters and Transmission parameters for the network shown below 6

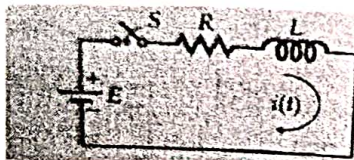


- (c) Discuss and derive the condition of symmetry and reciprocity in terms of Z parameters and H (Hybrid) parameters. 6

- Q. 4** (a) Obtain the Fourier series expansion in trigonometric form 1 and form 2, for the waveform shown in the Fig. below; 6



- (b) Explain and derive Initial value theorem, Final value theorem and linear convolution theorem in context to Laplace Transforms of time domain functions. 6
- (c) Determine the response of series R L circuit shown below for DC supply of E volt, using Laplace transform; 6



- Q. 5** (a) Discuss the properties of Positive Real Network Functions. Check the Positive realness of the function given by; 6

$$Z(s) = \frac{s^3 + 3s^2 + 9s + 3}{s^3 + 4s^2 + 7s + 9}$$

- (b) Synthesize the network using Cauer I and II form if function is $Y(s) = \frac{(s^2+1)(s^2+5)}{s(s^2+3)}$ 6

- (c) Synthesize the network using Foster I and II form if function is $Z(s) = \frac{(s^2+1)(s^2+4)}{s(s^2+2)}$ 6